

ActInfLab 2021 Quarterly Roundtable #2

Roundtable | 1:24:42

hello everyone

welcome to the actin flab and to our
2021

quarterly roundtable number two it's
june 1st

2021 and we're excited to be sharing
with you

some updates and some plans for our lab

if you're watching it live feel free to
ask questions or provide ideas in the
live chat or

afterwards you can do that in the
comments

welcome to the active inference lab

everybody we're a participatory
online lab that is communicating
learning and practicing applied active
inference

you can find us at our links here on
this slide

this is recorded in an archived live
stream so please provide us with
feedback so that we can improve on our
work

all backgrounds and perspectives are
welcome here

and we'll be following good and normal
video etiquette for live streams

today in the quarterly round table
number two we're going to

first start with a brief rehearsal of
some

of our lab goals and that will serve to
update

as well as to motivate then we will go

through each of the three organizational units of the lab

dot edu dot coms and dot tools

and be providing updates on what has happened over the last quarter

and about where we want to be going in the coming quarter

any thoughts uh at the outset or we can go right into the goals

so cool here are act and flap goals

for the one slide here we've just put up a screenshot of

our updated website because we want our website to be the point of reference and the sort of interface for

our lab to the external world so

our first line there is the goal of act lab is to produce cutting cutting-edge research

and enable real-world applications of active inference

and enabling architecture in the systems engineering

way of thinking about enabling systems can be

hardware it can be software it can be people and teams

so by thinking broadly about what actually

enables real world applications of active inference

it will hopefully help us stay open to enabling what needs to be

enabled and catalyzing what needs to be catalyzed steven

and then anyone else with a raised hand

yeah i think that point you make about what needs to be enabled is a good one

because there's this idea of
pushing beyond what's currently possible
if that makes sense in terms of
how these applications happen so i think
one thing that has been quite
interesting is that
practical applied desire
to apply active inference has been quite
fertile
so even if it's actually happening or
it's that edge of trying to make it
happen i think that's a good struggle
to be together in
and i think in tools we'll return to
talking a bit more about what those
applications
could be but i agree and the systems
engineering
approach of the enabling system at least
for me opened up to being focused on
what we want to
see happen while also being open-minded
to different things that could be
outcomes
and along the way of those outcomes any
enabling system
that needs to come into play is going to
be the right one so if we have to set up
a new tool or a new communication
mechanism
or do some other application it's all
part of the enabling architecture of the
lab
all right any comments on the website
activeinference.org overall or we can
just head right into
the updates cool because this is
hopefully an informative

roundtable lab meeting presentation so
here we come
to the updates first there's going to be
a general
admin or organizer update and then we're
going to go
into the organizational units
so first there's a special announcement
which is that
actinflab has a new steady state as a
non-profit organization
registered in california united states
of america
and the registration as a non-profit
organization
has been consistent with our lab
strategy to be a center of gravity
for education and open science research
related to active inference
and also increasingly incorporate
governance
and informational activities that are
related to our values
registering as a nonprofit will allow us
to enact
the kinds of policies that we expect to
see as consistent with ourself
while we also figure out what kind of a
lab we are and we want to see
the active inference community be as
participatory
accessible and productive as possible
so that's sort of a special announcement
on that front
and a big rationale or motivator behind
all of our activities as well as the
non-profit here
is this topic of research debt

so um alex maybe you could give us just

a

couple of minutes breakdown on what is
research debt and how does it apply

to active inference at this moment and

then anyone feel free to

ask a question

yeah thanks first i

saw that notion for research depth

in a blog post of jared timur

on in his syllabus for active inference

and he mentioned that he's trying to

work with research depth

and it's

very well accepted for

developers and developers

works a lot with their depth on

programming and usually

it's happening when people work with

some

new domain or some cutting edge

technologies

and trying to work on

new fields and developments and we just

not don't have enough time

to make some additional work on

bringing their developments

more closely to our people to have

understanding to start learning

it and so on and

definitely we understand that for

active inference as a new knowledge area

and for community a lot of

very bright minds working on

research part of it and

produced a lot of works and papers

and we should continue but

somebody should take with results

and try to

[Music]

to simply simplify it in some way to

maybe to

make it more clear and precise and

create some informational materials and

artifacts

for people who are not so

well developed in area and start just

starting

to learn active inference

that's why somebody just have to do it

and

in the lab we decided that it should be

our concern and we have

time and energy start with work and

actually i think we started it already

and what all what we are doing for now

in some sense related to this

big area so that's why we

decided to make it like explicit

statement about it like one of the goal

of the lab and for activities of the lab

so let's start think

and talk about it and think how we

should structure it

again in engineering way because

it's just a part of knowledge

life cycle and if you are working with

knowledge engineering it's a part

of knowledge life cycle and in our case

knowledge about active inference

cool thanks alex it just makes me think

of

um code that needs comments to be really

understandable and accessible that's a

problem

on the cutting edge if code is not being

applied properly to its fullest extent
or in a non-confusing way but it's also
a problem
for learners everybody other than the
cutting edge is also suffering from that
lack of commented code
and so we can just imagine if it was
just a script
that had one input and one output having
a function that didn't have an
explanation
would still be a hindrance to the
advanced performance as well as to
accessibility and onboarding
but we're talking about a rapidly moving
and evolving
body of theory and practice so it's even
more than just commenting
code it comes down to hopefully a lot of
the kinds of projects
that we're going to be exploring and
collaborating on
welcome blue anyone want to give a
thought on research debt or like what's
one
area of active inference that seems
especially
um underwater or is at the risk of going
into a debt mode
something we can think about and
hopefully draw out in this
round table and beyond like the
mathematical underpinnings
what is the relationship between free
energy principle
and active inference all these kinds of
questions
we can't just leave them unresolved we

might be able to
paint around them and make a clear
demarcation but that will be so that we
can
address it later stephen
did i ask a question as much as a
comment is could you just say a bit more
about
some of those concerns around
this research depth have been
overwhelming in some way
i get a sense that people might think
depth is just going to the most ultimate
abstract detail but we're talking about
a richness in a way i sense that you're
talking about a richness of
depth that can be accessible so maybe
just
explain a bit more about that be helpful
well
two similar sounding words like research
depth like how deep the field is we
would hope that
it's a field with depth and richness but
research debt is the hindrance that
prevents us from having
deep research and deep technology the
debt
is like if somebody puts in 100 hours of
work to learn something
and then it never gets distilled and
reintegrated
into the knowledge community it's kind
of like
we're building out very unsteady bridges
and so then there becomes 20 hours of
work to comment the code but then nobody
does that so somebody else spends 100

hours

and then they don't have the 20 hours to
comment the code and then someone spends
10 hours but it gets half done and never
gets completed

and so we accrue this debt just like
financial debt

about inexplicable models and partially
performed projects and just ways that
make the field

internally and externally incoherent
which prevents

deep research and deep thinking but we
want to approach it from

an engineering and almost like a sort of
information economy

perspective stephen so it's almost like
you know you have these bad instruments
or bad derivatives that they had in the
financial markets where they sort of
bundled up debts which

basically are non-productive in a way
and they just

so you're saying that type of thing
could happen there can be a lot of stuff
that can happen and it

ends up being not as productive and
just sort of jamming up the system when
it could actually be

leveraged and used potentially yep like
bloatware you know we won't call out any
operating systems or programs

but there's entire engineering
departments that are just focusing on
very low level bugs

and so then they're treading water
trying to fix the bugs or security flaws
in a really large piece of software and

that is a difficult struggle
whereas we want to be moving forward in
a way where we're not carrying and
multiplying our debts but maybe we could
even be
improving our returns and distributing
them informationally in an appropriate
way so research that it's worth the time
to unpack that because it is kind of a
core idea for us
the non-profit registration updates the
phenotype of our lab
and provides us with several new
affordances and also access
to a few new niches so this is
information that was
in our mailing list if you didn't see it
but we're better able to scaffold
what our lab does essentially and we'll
be able to interact with some new
opportunities
like grants and institutional
partnerships
and be able to provide a really credible
and durable onboarding interface
and at this point we're opening the
invitation
to participation in our lab from
individuals with various kinds of
experience especially related to
non-profit management or research
program management
grant applications the kind of legal and
financial services
that uh support all these operations
as well as a few perhaps new kinds of
things like
crypto economics could we make

non-fungible tokens
nft could we use a governance approach
could we use
quadratic funding this is something
we're exploring
on a few different domains as well as
people who can help with
graphic design or user interface tool
development
so hopefully if you're curious about any
of these areas and want to
contribute to the actin flab there's a
spot for you stephen
i was just wondering if you could speak
a little bit more and you might talk
about you know the
oh nft the uh ontology formal documents
narrative
and tools um which is you know
kind of i actually met that recently
against my
infusions framework which was quite
useful um for how that gets embodied
so i noticed it's quite um a practical
way to
to leverage knowledge in a you know an
engineering way
so i was wondering that that's a
particular
focus as opposed to for instance people
doing a focus on how to leverage stuff
for use in dance events you know for
that that may they may take a slightly
more of a focus on um
affect and kinesthetics and stuff like
that so
would you better speak to that if that's
okay

alex would you like to give a first
comment on onft or i'm happy to
please go ahead okay so in our
september 2020 paper with some of us on
this conversation
we proposed the onft model which is
ontologies narratives formal documents
and tools
onft or font works both ways and
the idea is those are different domains
that describe
especially remote teams where the formal
specification of the team
is kind of a almost perfect map for the
actual team dynamics
so if you have an ontology of
communication like there's video
audio text and file sharing then
you can have a structured system for
thinking about those pieces
that's ontology it becomes manifest
and used through formal documents
whether it's a
spreadsheet or a cloud document as well
as tools
which kind of covers the whole range of
what we work on on our computer whether
it's an interactive cloud notebook
or docx that's sort of in the tools and
formal documents domain
structured by ontology and then
narrative is what threads it all
together
and active inference hopefully provides
us with a way of thinking about
multi-scale narratives and holding space
for
differences in narrative and perspective

so then we can have a team where
individuals have a different narrative
sense of the ontology or they understand
different aspects of the ontology
or they understand a different meaning
for the different tools
but what coordinates them is the
relationship of the tools
the ideas and the people so that's kind
of how we're using onft
to hopefully keep the human in mind
during the design
and maybe even integrate it with those
kinds of performance
uh insights that are really important
elsewhere alex
yeah i want to add that
we was thinking about it also in
connection to
communication structure for remote teams
and trying to find the ways how to
increase
efficiency of such communications for
people
with which is different languages which
have different cultural backgrounds and
all that
different difficulties that
that we meet when we start working
together
and we have that notion in the beginning
about uh
communication nightmare for especially
for
actually it's happening even in classic
form like in corporations
in big companies people trying to
discuss something but

they use they might use the same words
but just understand different by under
that words
and we was trying to find some object
that we can like make more formal
to and make explicit to think about
and maybe we will add something later
but for now that ontology formal
documents
those narratives looks like
basic object that we team need to pay
attention for
yep and it helps us generalize beyond
the language that we're using to deliver
a formal document
in so then adding a translation on a
cloud document or adding
text to speech or speech to text those
are not bolted on
second layers it's part of the delivery
mechanism for the information that
everybody
should be receiving in a given project
so we can incorporate
accessibility and differences in
people's tech setup into a fundamental
of the team
rather than imagining that there's some
single style of
default team member and then we're going
to have correction terms or something
like that
so here is a graphic of a few of the
parts of the lab and ecosystem we're
going to now go into
internal to the lab in q2 and
in q1 we have three organizational units
which are

dot edu education dot coms
communications and dot tools
and you'll see that dot coms
is having an interface with a live
stream like the one
right now which is the production of the
comms unit as well as comms
now and in the future potentially could
have interfaces
with broader communities adjacent
communities
um whether through the participants in
those communities like the blue square
on the right or whether to organizers
who then are able to participate in the
lab or somehow integrate active
inference into how they think about
their work
the live stream interfaces publicly so
that is
the external niche to the lab and that
includes
those who are learners and practitioners
of active inference
self-identified or otherwise as well as
those who are not
self-identified as active inference
learners and practitioners
edu also has an interface with a body of
knowledge
which we'll return to but that's an
ongoing project that will include
multiple
components and that's drawn from the
systems engineering
body of knowledge concept alex and then
anyone else
yeah this notion and with graphics for

interfaces it's not only from system engineering but also we was trying to apply it uh concepts and understanding from active inference and to show that boundaries for us as a lab and have an understanding that such lab should have some interfaces to environment and more but as we understand it for example uh interfaces should be connected to communication units so tools for example or edu shouldn't have external connections just to have that consistence on the lab scale to understanding how that communication how it interfaces work and how we can manage it in most efficient way cool any other thoughts here otherwise stephen and then we'll head into the units well just to comment i like also you refer to niche there and i think talking about niches and we talk about systems we've talked about system boundaries and where mike markov blanket type interpretations been used so i think that way of thinking is um we're kind of getting used to it but actually it's kind of quite unusual in a way there is some work with niches in um regimes some some socio technical system work does use

regime but this ability to move
into thinking about things as a niche
types of niche
types of system having an internal state
that's
actually got an agent involved as well
at a smaller scale
and i think we're wrestling with that a
little bit exactly where that all lands
but i just thought i'd
notice because you use that term that's
been quite interesting
good point thanks yep we agree
thinking about interfaces is natural to
computer science
the idea of an api as well as to active
inference
so it's kind of like a bridge idea
between those two
ontologies or perspectives and
now we're seeing an increasing amount of
work from the computational the
ecological
all these different angles that are
coming together that
are describing systems the way that you
just laid out
okay so dot edu
is our first organizational unit
the goal of educationunit.edu
is to create a participatory
and dynamic active inference body of
knowledge so the body of knowledge
which we've described more
thoroughly in i think probably our first
round table
the body of knowledge is an idea drawn
from the systems engineering body of

knowledge and it includes
various kinds of products and enabling
architecture
all defined through interfaces and
relationships
with an ontology like this group creates
this one or this one consumes this one
pays for this one is paid by that one
and that allows us to have seemingly
disparate kinds of
units in the body of knowledge like
certification
practices competency training of course
educational
material and curriculum all these kinds
of
structures can exist within a broader
network
and we can identify where is something
needed where something redundant
where is something needing to be tweaked
and as far as our progress or actions on
that front
we have developed the ontology
a little bit which we'll walk through
and we'll also mention the ontology
working group
we've released our terms list v1 and
we're in the process of
updating terms to v2 with an increasing
number of language
translations and we'll cover the next
steps in a second
so here on 14 and 15
we're just showing a snapshot of
this document which everybody is welcome
to view
and comment on this is the first

tab of our working ontology document
and the first tab has the core terms
in green followed by their references
up to five per term at this point
and citable definitions where the term
is used

so not just papers where it's relevant
but actually where we can point to
and say the term was used in this way or
defined this way by these authors
at that time so this has been a big
literature

review effort and that's been quite fun
then on the second slide we have those
same exact

core terms and there's the introduction
of the columns reflecting
natural human language translations so
at this point

with the help of many on this call
we have russian portuguese spanish
french and italian all to reasonable
degree of completion and other ideas for
how we could

improve the state of those languages as
well as

introduce other languages and symbol
systems so we also have a supplementary
terms list which is sort of our holding
area for words that are

relevant but maybe not in the core
and one thing that we're looking to is
the kinds of governance mechanisms and
collective decision-making mechanisms
that would help us understand when to
elevate a term from the supplement
to the core and vice versa so any
comments on

just the terms translations references
definitions

alex i just want to share news
from yesterday that we have full
completion for portuguese and
russian nice crucial languages
um let's look at where this

ontology working document is in our
broader

past present and future so here
we have uh drawn from a figure a systems
engineering

dynamic life cycle ontology paper
from the left to the right there's a
continuum or a spectrum

of levels of formality of ontology
so the entire left side you can see is
defined as informal and lightweight
ontologies

whereas increasingly expressive and
formal ontologies and ontological
systems are towards the right

so in the q1 roundtable of 2021

what we had was a list of core and
supplemental english terms

so we were just then deciding what what
terms

should we make the first pass for core
terms

since then we have updated the term
membership we've added

terms as well as we've

started to think about how we can
clarify the distinction between core and
supplement

and what you've seen in these previous
two slides were the introduction of
references and definitions

which is as stated right now it's a literature quest but eventually we'll be looking to synthesize and integrate definitions and be able to say here is two senses of this term or this paper is using it in this sense or even in this sentence it's used in this way and then in the next sentence it's used in a different way we've also introduced and really improved the translations so that's something where everybody is willing uh who's willing to would be welcome to contribute and then we've continued our march from the terms to the blue highlighted boxes which are web directories the references are a web directory they point to links on the web user classifications into the core and supplement data dictionaries and structured glossaries which are describing how definitions were used in papers and then we have this sort of pink line that's written as expert help and ontology working group and then here's ontology working group like a train kind of pushing to the right and in the ontology working group where we're working through this awesome book by adam peace ontology um we're going to be making that sort of two hands reaching out to each other so as we

improve

our ontological competence we're also

going to be

increasingly engaging with experts and

so

experts are welcome to assist whatever

their

level of interest is in active inference

and over the coming months

we're going to be working out in the

ontology working group

how we can be making an active inference

ontology

hello again yvonne

any other comments on this ontology

slide overall otherwise we can just

close and just say everyone's welcome to

get involved with project-based learning

and broadly this is knowledge

engineering

we're learning by doing and we're

learning by doing by engineering

and it's about language so especially

people who are interested in language

translations

if they see a language and they have a

friend or they know a language

that could be added there's just

millions to billions of people

who don't have the english skill per se

to

read the research literature so we can

reach out to them

and one way to do that is through

translation

we can explore definitions and

references better so for those who like

searching the literature for how things

were
said and meant edu will
as part of the body of knowledge focus
on curriculum developments
which is the structuring of those core
ideas
into curriculum that makes sense
and in our ontology working group where
we have a total
range of skills but we're all learning
by doing here and we're trying to
integrate best practices from the
ontology life cycle
models of systems engineering as well as
ontology developments
from sumo for example starting with a
list of core terms
and a narrative that concisely describes
the space
steven and then anyone else and also
comment how
it's actually been very useful from a
knowledge synthesis
just a learning perspective to go
through the terms
and see how they fit together and
get a bigger picture on what can be
quite overwhelming if you're just
reading papers
so things like what are
action states what are sensory states
and then you start to see there's a
number of these types of what's it to be
a non-equilibrium steady state so those
questions
are starting to sit in the background
we're holding back from getting too
sucked in by them

but there's certain patterns i think
that have come up with the
terms which has been quite useful
especially having done these live
streams with different people over the
last
year and seeing how some of those same
um ontological questions have come up so
i just thought i'd mention that thanks
uh yvonne did you want to add something
there yep go for it
yeah thanks i just want to
mention that this schema is quite
uh quite linear and it has
a lot of checkpoints and
during this quarter we made uh
for six i guess
six points and
if we have
we will have the same
the same speed
might be we did
we will we will make all all
this ontological work but but i think
now we uh just
from starting point and we have
a lot of
difficulty and interesting work within
this
anthology working group
thanks totally agree this um shouldn't
be understood as us being 40
done even when we're way way on the
right side
we still won't even be a little bit done
because there's so much to do
and also it's so true that whatever
level of skill we entered into

this ontology working group with we've
all learned about domains that
we just weren't working on so we weren't
on projects that involved
translation and just the issues that
were raised
regarding the subtlety of translation
and about how terms were used through
different
epochs in the literature these are
insights that just cannot be
achieved outside of learning by doing
there's no one who is waiting to tell us
that information and then we called them
this was like team working and finding
out cool
things about active inference
okay so on to comms
so this live stream a comms presentation
the goal of comms is to organize the
labs
internal projects and activities and
that's driven by an active inference and
communication
oriented worldview so for example
um a world unto themselves active
inference and communication one of the
early papers we read we started to see
the organization of teams especially
remote teams
as being related to the structure and
the quality of communication
so just like the capacity of the thing
on your desk to be a computer
is related to how the wires are
connected which is a communications
network
the structure or the capacity of a

social system

is related to the communication patterns

that it is using

at rest or in activity or

during realignment so communication

engineering is a key principle that

integrates what's happening

inside of the computer to between

computers

to the human in the computer the human

and the human computer in the middle

all these combinations are subsets of

broader ways of thinking about

communications architecture

so that's how we think about

communications internally and then

we of course think about communication

on the interface between

our lab and external stakeholders

because comms is in charge of carrying

out all the forms of communication with

external entities so comms is the one

who picks up the phone when you email

activeinference gmail.com it is the

interface that we have

and then information can be passed to

other

organizational units or roles as needed

our progress on this front was we

continue the podcast and blue you're

welcome to add anything

about the podcast we also continued

and developed with the help of

especially dean

and steven and blue on the live stream

where the dot zero video is becoming

contextualizing and a lot more it helps

us

learn the paper and prepare for the
discussions and that's really been
allowing us
in the point one and the point two along
with authors who
in almost every case have just been
extremely generous with their time
and in their interactions with our lab
it's helped the dot one and dot two
be real jumping off points like hop skip
and a jump
so a triple jump with the three sections
rather than just
logging through the content three times
the same way
and we'll talk more about it in a second
but we also continued several of our
streams um blue what would you like to
add there
yeah just about the podcast i think we
were just launching it
um in the very like in the last
quarterly roundtable like maybe had one
episode out or maybe not but we've been
trying to release them
weekly still trying to get interface
with itunes but
it's up on android uh google podcast
um and one there's one more platform
that it's on but it should be available
and it's available through
the rss feed so there's a like
sequential episodes that are already
ready just backlogged i'm just releasing
them weekly to try to make it
like more regular as i remember to
release some weekly like
some i'm trying to do it on the weekend

but sometimes it's like tuesday like oh
no i forgot
uh or it's like i'm going out of town
this weekend so i'll release it early
but
um hopefully that'll be automated at one
point
also so just update there cool
and the podcast is awesome for those who
haven't listened
it takes audio sections from live
streams
earlier ones maybe that you haven't
listened to and
it slightly recombines them and provides
introductory context
that makes it a lot more direct
listening
and so to pick out the the trail to walk
from just the map is an awesome skill
and so thanks blue again for that
yeah it's also just good for people who
are maybe not looking at a computer you
know i mean it's
like if you're it's it's not an attempt
to really get into the technical
um like dive into active inference so
it's very
much just like an interesting discussion
because like even though
these concepts can be very technical the
fvp and active inference and there's
lots of
equations in math going on and we are
going through a paper this is like
you can step away from the paper and and
just listen to like the interesting
conversations

that have been sparked by the paper
which that's kind of cool too to just
see how people approach it and how they
think about it
and also um in many of the episodes
somebody's introduction
is placed before the first time they
speak so
like a question will be asked and then
it will go to the introduction which
probably happened
20 minutes earlier in the stream and
then someone says i'm in
xyz area working on this topic and what
i think about culture and communication
and then it's like
ah that connected the dots because that
was very linearly distant
and during the stream listening to it it
doesn't always pop out that same way
steven and then or blue and then yeah
yeah let me just say one more thing
so i am working on um
like smoothing out the audio because we
all have different microphones and as
like i'm cutting
different like excerpts out sometimes
the audio is not like smooth so if
anybody has any resources
or skill like with audacity maybe i'm
doing what i can with obs
and the filters in there but um if
anybody has any pointers on how to
smooth out the audio like i'm so happy
to
uh take that advice on thanks
stephen yeah good
good good point i i just wanted to add

that i think it's really useful for
being able to go to just certain bits
and re-listen to them
because often it's really good to see
the or to see or go back and see the
whole of the um
live stream but sometimes the useful bit
is there's a certain
20 seconds and sometimes it's useful
just to listen again and again and it's
like oh i get it now you know or
because it's just too much sometimes the
first time you listen to it and
you know sometimes people probably see
our eyes going a little bit panicky when
it starts
trying to track what exactly has just
been said so sometimes though having the
chance to
yeah listen to it a few times because
once you get one bit then you start to
unlock
the other pieces start to make sense and
that that's quite useful so
so yeah thanks for doing that blue it's
really helpful
yep it's like a many coats of paint on a
really big
mural so okay communication we're
thinking about it
this way instead of this other way okay
got it and then you hear communication
in another context so
we add to the depth of our understanding
we as we link
ideas and see people who have come to
the ideas from a lot of different
angles here is some of our

live streams that are just screenshotted out

so we're up to almost 60

after we do 23.1 and 2 in the coming weeks

almost 60 classical live streams

that's the regular tuesdays session

we're up to seven completed guest streams

so in the past quarter we've had some excellent discussion on the technical side and also on the technical critique side which is very important for the field in five guest stream

from martin beale and then in guest stream seven

with miguel aguilara and others and also

with odyssey unica heard about

co-embodiment and about first prior so expanding the gamut and those guest streams are

are open invitation to anybody who wants to present on something that's

relevant to be brought to the lab and

hopefully we'll continue to show that

there's many ways to do a guest stream in the model streaming just a couple of days ago

we had uh ozone catal and tim verbellen talking about robotics and that was some of our first foray

into robotics and it was just awesome

and inspiring to see a little robot in the aisle

and to understand how all of these different sensors could be

integrated on the fly and then to hear their distinction between reinforcement

learning and active inference
so that's some awesome engineering work
and it's exciting
we also had a model stream 2 with noor
sajid and philip
ball also extremely helpful
from their demystifying active inference
paper so
cool streams there we had six streams
with john boyk four background streams
where we went through on a jamboard his
tripartite
paper on sustainability and social
systems
change and science-driven transformation
so john just
really took it to the next level with
with an author
engaging with us before the guest
appearances
and then we were able to have a dot one
and a dot two that was also
really interesting and then on the
bottom left there
we've had two math streams both with
shawna dobson
and then we'll have another one with
more of a group
discussion with blue and shauna and
others
next week and so math stream again just
there's so many cool ideas that come
into play
like dynamical systems or control theory
category theory those are big topics
you know you could have a whole section
of a bookstore or a whole
major on those kinds of topics and

math is a big area math is the
generalization layer
for science in a lot of ways so we can
work towards
each of these series being able to
raise up different aspects of active
inference
as well as be like a two directional
highway
connecting those who are interested in
active inference to ideas that they
might not have considered or known about
and also people who are considering
those ideas every single day
bringing them a new perspective on
active inference
so yvonne and then stephen
yeah thank you i just uh want to
totally recommend for all who
just starts to deep the active
inference uh to
watch uh dots uh
zero streams with uh
uh with our colleagues then dean and
blue
and uh say
thank thank you guys it's uh
it's totally helpful streams and
uh for me now it's preferable
type of watching streams it's uh
just uh online youtube because when i
watch it
online with the room with
people inside the room i
often find myself lost so
when i when i see it on youtube i have a
chance
to go back and after that just

on doubles double speed to
get it all almost uh
online thank you yvonne
stephen and then anyone else maybe dean
if you have a thought on the
dot zeros as well but stephen yeah
those dot zeros are really helpful and
it's been really really
exciting to have these real specialists
in math
so he said this we think that active
inference has got a lot of math but then
these are the people that are more into
the math than even the active inference
that the active inference people are
often talking to
you know in some and to have them given
a voice and i think they really
appreciated that but to have a voice and
to be questioned
and to be brought back the other way i
think it's really helpful because i
think
you see there's a lot of people coming
attacking
the word attacking is i'm not being too
dramatic with that but
attacking active influence from a math
formalism perspective
and so to have it just coming from
people
and they talk about their own area their
own fields of math which might have
stuff which isn't so clear they're
questioning
they're trying to explore particularly
category theory stuff
right on the edge i think that's really

really valuable

um and i think that's going to be

something that i haven't seen anywhere

else

so i think that's really great yep it's

been really fun

so anyone else want to add a comment on

just the

live stream um repertoire or diversity

dean

and then anyone else

yeah thanks ivan and thank you to steven

and blue and

daniel and anybody else that's done

those um

point zeros um so

i i just wanna talk for one second about

the getting lost piece i think that's

something that should i know it doesn't

sound like a doesn't sound fun who likes

to be disoriented

um and if you are if that's kind of

personality you are the disorientation

is your is your drip i guess then that's

that's cool but for most people they

want to kind of get the

control back today being a perfect

example i didn't have

the slides i had to figure out how to

get the youtube on a second

window and turn the sound off and so

i was dealing with my own markup blanket

for the first

25 minutes of this and i think

one of the things about the point zero

is and i i think it really matters

is that nobody wants to be

serially disoriented people want to be

able to get into that place where they find themselves reoriented and i think however we go forward with the point zeros we always want to kind of keep that

front of mind we want to really be explicit about the fact that at any given moment uh your ability to sort of keep up

i keep talking about this all the time the ability to sort of keep up with the rate

at which this information is unfolding isn't you're not always going to be able to keep your optimal grip or hold the butterfly or insert metaphor here thankfully we have daniel who's really good at metaphors um but most of the time if you can't bring it off the abstraction shelf and put it into some sort of material sense of something that you're familiar with you're

constantly feeling like you're catching up and so

i know there's going to be a lot of time spent on getting some sort of a standardization process and i

i know i understand that you're trying to create

an engineering piece that's not too instructional and his more interactionist and all that that stuff and i think that's that's happening but i think at the base of it it's just just a real empathy for the fact that people

who do join i joined this part way

through this quarter
so i'm like a two-thirds of a quarter
kind of guy right now that's why i
kept my mouth shut um but i think we
have that empathy for people
who jump into this regardless of what
their experience level is um i just
think that that's um
i think that's going to be really good
in terms of maintaining and
maybe building on this in terms of more
people feeling comfortable with
like me i don't even like to open my
mouth in these situations but
you guys have pulled it out of me so i'm
just trying to figure out why that
happened and i think it's because
there's a certain kindness and empathy
thanks dean yvonne
i think all time long the
lap will be exist
every every time will be uh
people who just started to
try and understand active influence and
and in this case where we will be
in in the same point
as as we now
as we are now thanks
blue so just in response to
to dean i'm i'm like way farther in
right maybe like
i don't know nine months or something
into the active inference study and
um i still feel like i'm catching up so
i think it has to do with your
background there's like a relational
framework
we're all coming from different

perspectives some people are well-versed
in philosophy or mathematics
or for me it's neuroscience so whatever
your background is like you kind of can
fit into the active inference from that
angle but then like
you're still trying to catch up in every
other angle like i don't know anything
about philosophy so it's all like new to
me um and some of like the
like i'm good with math but not like the
formalisms of math and how like
you know there's lots of symbols in
there i'm like what does that even mean
um so you know i think we're all kind of
approaching it from a different space
i'm still
i still feel like i'm catching up all
the time so i don't know if that feeling
is ever going to go away i
i kind of hope not
nice totally agreed it's not a uh
finished
skyscraper it's not like everybody who
has caught up is waiting on the 100th
floor
just laughing or something like that
it's like a diffusion of an
ink spill and so is any
ink molecule catching up with what the
edge
well if they want to they could probably
get there in time
and um those questions what is active
inference and how do we apply it just
like what is complexity
how do we apply complexity those are the
questions

that bring in an introductory person and also remind those who have been in the area a long time to have a beginner's mind so having a question that can be understood on multiple levels is really important for integrating the community because it's not like there's one question that gets you in the door and then we switch out the question for another question later we can be working as an integrated whole on what active inference is clarifying that for ourselves and through our um constructed niche stephen yeah i think that also dean makes quite a good point that the new affordance of the dot zero to help that orientation process it does help because often we're bringing someone in for the first time to meet us to share a paper and we may have slides we've prepared they may want to suddenly bring in some other slides because a couple of days before they suddenly had a synthesis themselves so having that dot zero is quite a nice way to sort of like you had in mind there as as a way to orientate people and maybe even feed a couple of um bones so to speak for people to chew on for the for their journey even if it doesn't go into all the details so maybe we'll learn some strategies or

how dot zeros work and how to
become more expert at using those
opportunities
yep we hopefully show the authors and
our
participatory audience that we're taking
it seriously
we haven't just asked them three months
ago to show up at a random link and then
we didn't think about it before then
it's also the work that we have to do to
have the slides and the fluidity
ready and it does so many things and
dot zero the prerequisite for
participating isn't understanding the
paper
you'll find there was papers where we
just traced out the section titles and
said okay
we want to learn about that we want to
learn about that we're curious about
this
so those who can um
come in a live stream and be curious or
can assist in the slides if they don't
want to come on
live that could include every single
background
because we're not evaluating on a rubric
of how tall are you on the active
inference understanding scale which
doesn't even exist at this point
so cool about the dot zeros let's talk
about
another production of comms that will be
the karl fristin mini symposium on june
21st
2021 so this is going to be and anyone's

welcome to raise their hand and give a
thought i'll just give kind of a
logistical overview carl has written to
us
and said um in fancy italics
i generally find these kinds of sessions
most productive
if we are i am in response mode and deal
with questions that people bring to the
table
so to that end we've decided to
structure the symposium
along three sessions each driven by
one of our organizational unit and then
we've also
been soliciting and compiling and
integrating
a lot of awesome questions so everybody
here has helped uh generate questions
and maria thanks so much for
also helping on a few other aspects
for anybody who wants to get involved
there's a google form they can fill out
if they want to stay updated when the
videos are released
it won't be live streamed but we'll
release the videos shortly after
and it's an opportunity if somebody
wants to co-organize with us
on this event and then be able to
participate in the proceedings live
that's totally an option and
yes we're mentioning it on this live
stream
because we want to solicit the kinds of
questions
and foci for this event that would be
most becoming of the special opportunity

that it is
so anyone want to add a thought
on what might be exciting about this
carl first in symposium it's also a
little bit like our end of
semester um moment
so it'll be a amazing first semester
and then we'll pull back for a few weeks
continuing
our operations and learning but
preparing for the next semester to begin
in august
but it will be an awesome way to end the
semester stephen and then
yvonne and then anyone else yeah well
i'm hoping
carl fristan be really interested by the
interdisciplinarity or trans
disciplinarity of the uh
the participants that are in this mix
that we've been sort of
sort of inside and exploring so
i'm really curious how um
we can bring those kind of
applied questions to the table um
and keep it all within um sort of a
manageable
scope because uh we're seeing so many
things that could come up
right and there's there's a lot of huge
areas so
i think as we start to refine the vision
of active inference lab that also is
happening
is helping a little bit give him an
anchor to where we're coming from so
um i'm looking forward to this next
couple of weeks as it starts to

shape up totally agreed thank you
uh yvonne and then if anyone else wants
to add a comment
yeah thank you uh
to meet uh carl uh
here in lab it's a very
big pleasure big honor and saying
thanks for all participants and for all
guests
that we have carlos attention
and for a community as well
as like other
other community we we can have some
special
special event for one one year to
have carl one once a year it will be
like like our holiday
cool thank you dean
one of the things i want my
participating in this is to get
hopefully some more insights because i
did a lot of programming
using active inference before
i joined this and
most of the time my colleagues i
couldn't i couldn't even broach the idea
of active inference with them because
they were so this was so in
their opinion this was so out there that
it wasn't even as
accessible to them and i understand that
this process here is to
try and build a foundation and one of
the things i'd like to talk to dr
furstén about is
so when does this
when does this become more available to
the general masses because i i truly

believe

you know based on the conversation of
the last live stream

there's still things that are being done
every single day

just because that's the way we've always
done it and i think what he's
introduced is some new ways of looking
at things not to discard everything that
was

ever been done because that's not very
respectful but

i would really like to know how you can
make this accessible so that when you're
sitting in your backyard

having some pops with some friends you
you

don't even want to talk about it because
even though you find it super exciting
um

they look at you like you just grew a
third eye and

what i just came here to relax right
because i think part of this

for a lot of people is um

it sounds really really cool and
interesting but it also sounds like you
need

erickson's 10 000 hours just to begin
looking at it with any degree of
sentience

so talking to dr firsten about how we
make that

that ant hill a little bit more

accessible would be really cool

cool for some reason sitting in the
backyard it just made me think of

some friends they're watching a bird are

they going to understand the birds
behavior

in terms of reward maximization or will
they understand the bird's behavior
in terms of reducing its uncertainty
about its

generative model and its niche and the
evolution of birds

or will it be framed in the economic
quote rational agent reinforcement
learning framework

so the shift could take place in the
backyard as well

stephen yeah and i think that's a good
point

that uh those different ways of seeing
things and of course active inference
in application it moves to approximate
each of those at different times so i
think that's what's quite interesting is
um the general desire to give someone
the answer

is so ingrained in our culture because
we happen to think we've got the answer
to everything

at some subliminal level so we're we're
a little bit trapped

and we all are to some extent in that
because it's so ingrained even though we
try not to be

and therefore this idea

that this question about you know
what's this more like

gets gets um gets into that so maybe
having

some different types of work around the
types of metaphors the types of examples
he has that classic one with the cat

that looks like a the cat's ears that
which in the shadow looks like a wolf
howling right
and then you realize you know but at the
end of the day it did the job
so you know there may be some of these
questions around
the sort of what would we say about this
in the back garden could be actually
quite useful because i've had this very
similar
trouble in many different ways and you
know
sometimes you just wish i hadn't brought
it up you know because
but that that is changing now
yep fun stuff um okay and then
just anyone raise their hand if they
want to add anything else oh yeah alex
go for it
yeah thanks i just want uh to add
shortly
uh we had the red conclusion in some
discussion that
person need to be active inferent
before he started to learn
active influence right if someone the
first question
is how is this going to increase my
paycheck tomorrow
you know that's easy to answer it won't
but
for those who are wanting to reduce
their uncertainty and understanding that
exploratory play you know early on
playing with a toy just like i'm playing
with now
it's exploratory and then maybe i'll

make something rewarding with it later
or maybe not
maybe i never need to turn the dial that
way and so people who
approach the conceptual world
with that kind of a spirit um
with play and reward on a sort of
fine-tuned
balance and a sensitivity for
uncommon patterns and just being
in relationship with the material i
think that they'll always go far in
active inference and by doing so they'll
push the whole field ahead
so one other piece here is just the
our live stream checklist which we kind
of
increasingly hope that individuals
follow before they
go on a live stream ranging from just
things that they can do
ergonomically to improving their
internet bandwidth and their cognitive
bandwidth
these are things that could hopefully
help us propagate
good practice for online team
communication
in live streams at first where the
performance is
of the utmost but also in other
situations like
thinking about video chats that you're
in as respecting your partner's time and
requiring uh preparation and then we'll
just close this section with the
affordances for participating in dot
coms blue go for it

so just from the i am evaluating
the quality of your audio um side uh
just something to put in there in the
facilitator
checklist like having my my shiny new
condenser microphone which i know some
of us have
it's like life changing in terms of how
audio quality sounds so if you are
replaying the
um the live stream videos and you're
wondering like
why do i sound like crap well it's
because your microphone is crap
i just want to put that out there it's a
small investment like 50 bucks i know it
can be like
life bank breaking if you're a college
student sometimes but
um you know i really just as someone
who's evaluating the quality of your
audio
do us a favor audio engineer blue
saying it so that others don't have to
anyone who
is a repeat participant will provide
them the hardware to participate
and that is going to be fun i think as
we continue to explore
hardware and software affordances for
participants
maybe a kit do you need just the
microphone but your internet is good do
you need just
a connection to the internet there will
be participants who will be able to
contribute
with that enabling architecture

those who want to participate in dot
coms

well there's several things that can be
done first

anyone is welcome to get involved with
contributing to the current series that
we have

whether the podcast which has been
mostly blue's effort

or any of these series that we have like
the live stream paper discussions
the guest stream model stream math
stream so

for these formats it's possible to help
us

co-organize make the connection to
somebody who you think should come onto
a stream or help structure a
collaboration in another way

but if you saw one of those videos and
you liked it or you wished that there
could have been one

like another episode of the model stream
with a different robotics lab
or another math stream on a different
math topic that's the right way to go
about it

and also it's possible to introduce a
new series

or format maybe we'll have so many
robotics discussions

we'll have a robo stream or we'll have
enough

miramikology research to have an ant
stream

we can also focus on onboarding
new participants to live streams so
people who've come on once and kind of

thought it was fun maybe they want to
help others do the same
we also want to explore other
communications
format and media like events
uh series like a discussion series could
be recorded
or not you know what stephen just
brought up with the metaphors or
you know making active inference
tangible you know active inference in
your hands
those are kinds of things that could be
a fun different series
also there's different formats blogs
short writings tutorials
meme generation of course always a job
opening for meme engineering
um but dot coms is the interface with
the external
so anybody who likes that sort of
communication oriented
work is going to find a home in comms
and also
comms plays a role in the internal
structuring so for those who want to
understand a little bit about how
communications engineering internally is
used
for an efficient and inclusive and
accessible teams
that's a different environment operating
than to the external but both kinds of
communications are really important
okay going on to tools anyone just
raise their hand if they have any
questions and the live chat people can
ask if they have any

comments or questions all right so here
we are in our third
organizational unit of tools
the goal of tools is to enable effective
tool
and instrument use for everything that
is happening in our lab
the tools organizational unit explores
and designs affordances for our niche
which will result in effective action as
well as innovations
and probably product developments that
are downstream of working on the areas
that we see
as important there's been awesome
progress related to weekly meetings for
sharing resource
needs and ideas as well as a series
of brainstorming and development product
development uh meetings which i'll
leave anyone else to describe and the
current next steps
are focusing on onboarding participant
into the tools operations
understanding the lab tool use and needs
as well as some other
outcomes of again this series that was
undertaken by
alex and others so alex do you want to
describe what
transpired there or
with the tools um yeah yeah just
i want to the comment and with
connection
for uh needs
from blue that she mentioned so
and it will be like an example for what
the tools will will do so

bloom mentioned about some needs for automation for podcast and also some maybe tools for smooth-out audios so we will need to have some maybe special session or just make it a synchronous to document such needs and then dot tools as enabling system to support uh technical parts of our organizational units jobs we will work on it and provide some solution to needs for working with podcasts and so it's for all our possible needs from organizational units what we have now or will be in future this entity will support them in technical needs yes yvonne i just wanted to add that there is no main unity in the lab so all three units are like uh each scuffles the other and from auto enabling system that every every so every one is the is the main and uh every is and issues has a special special program yeah to to help with you know others thanks great point it's like organs with specialized function you can't have one organ that does it all and so they co-enable each other in the context of a broader whole

this is a figure from our september 2020
paper
active inference and behavior
engineering for teams
and this was our representation
of thinking about how teams were linked
through basically informational blankets
and so inside of the team blanket the
team markov blanket
you have team members who are bounded by
their
human computer interface and they are
able to pass information back and forth
to databases single source of truths
whether it's the jitsi server
that all of our video information is
relayed to before it gets
re broadcast back out to us or whether
team member b
is watching the live stream and
youtube's database is the one
that is taking in my video streaming and
outputting
on their screen video however we want
to model this this is kind of our big
picture for thinking about for online
teams
what kinds of tools and systems are we
engineering
um steven yeah i like how you've got the
database sort of
inside the team mark off blanket as well
so it sort of shows how you can have
parts of the niche on the external world
on the inside or the outside of another
scale of blanket um so
you know that type of um
way of interacting that's not so common

with other systems approaches
so this this nestedness of there's no
definitive inside outside
as the scaling blends um so i
you know i just thought i'd mention that
yep cool
and here's how we hierarchically nest
in our google drive our cloud storage
folder
we have an admin folder for organizer
information
and then each organizational unit has
information that's accessible to it and
so in a
unit and project project-based way we
have nesting
of reading and writing privileges
that allows us to sort of unambiguously
understand
what information does somebody need to
see and then what information
are they contributing on yvonne do you
still have your hand raised
okay so that's our google drive just to
sort of walk through a couple of our
other tools
on the right we have a lot of google
documents and sheets this is our formal
documents so we use
various kinds like slides and sheets and
docs
on the ontology tools side we a little
bit
have used the online although there's a
downloadable as well
ontology editor protege and then also
we've been with some help from tim and
others

working on setting up a sigma ontology
development environment
which is related to sumo the ontology
approach that we're
pursuing so this is a
formal approach for ontology modeling
and a
example of a of a statement in pseudo
kiff
is written here so that's what the
ontology working group
is developing and it shows again the
power of working on a team
because how many of us could have ever
set up the cloud
uh sumo environment none
or maybe one how many could even set it
up locally again
none or maybe even a few but we do have
somebody who has experience with cloud
setups
so we will have a cloud working space
and that is going to enable participants
who want to
have that ramp from not having sigma
not knowing ontology to maybe having
both and understanding both
it's going to be an easier and a better
ramp when we can share resources in our
team especially the
sigmontology development environment
still use key base as our back end and
single source of truth for organizing
although we we have a discord as well
that has been awesome for
a lot of collaboration for communicating
externally
our primary source is our monthly

newsletter

which sometimes sends out other
announcements as well and then we
have a discord with people who want to
be in a more

uh web platform interface it's been
great for

the collective intelligence project that
blue and several others
have been involved in and then twitter
we've

used that social media affordance as
it's where a lot of the community is
then on the audio visual side this is
sort of all down onto one slide
um using youtube for live streaming and
viewing

using jitsi and gather town for video
chat we're on a jitsi
right now and then using obs broadcaster
role

okay any comments on tools otherwise
we'll head to

future and the next steps all right so
hashtag future and really our next steps
as we are heading into our third quarter
of activity

we first off are going to be taking the
next steps that we described in each of
those organizational units

so edu comms and tools we'll continue
navigating uh wayfinding sense making
along

the ways that they've been describing in
this conversation

as the lab as a whole heads into the
third quarter of activity though
we will be focusing more on defining

governance structures
and exploring new affordances that are
now available to the non-profit
which was part of our strategy from the
beginning achieve the non-profit
understand which affordances would now
be possible
or even likely and then enact those
possibilities
so the lab level that's what we would
like to see and also
experimenting and defining governance
structures
whether it's a governance structure that
we don't need any new
technology or tools for like just
clarifying how do we
make a vote on whether a certain guest
is appropriate for a guest stream or not
all the way up to the token engineering
of
nfts and more defy or distributed
governance
type applications for who knows what the
next steps may be
and then also during june it'll be an
awesome
note to end the semester on and then
we'll have a third and a fourth round
table
before the year ends well
we can be on this closing slide which is
just
anyone is we can open it up i see dean
and then everyone else feel welcome to
raise your hands
dean go for it yeah just a quick
question i

i noticed that the terms of
terms of whatever they call it on
youtube changed and now
when i've gone back to some of the
active inference live streams
before there was no advertisements and
now i have to click through two
advertisements for example
to um actually access the live stream so
would we get somebody to look at the
adsense piece and start if it's a
non-profit start
collecting some money for people because
the advertisements have been
actually front end loaded on some of
those live streams
you probably weren't aware of it because
it's just happened in the last
i think 10 or 12 days
thanks i'll give a thought on that blue
yeah just to respond to that so like we
as the active inference lab like
can't we can't pay to
remove the ads it's a user end payment
which i
that that's my understanding of it so
just a quick update
yeah i think there is a way to like have
a channel
under monetized or not i'm not sure if
it removes all the ads
uh but yes that's what changed
right so even if you're that's exactly
what changed
there used to be like you could pay as a
producer
so that no one had to view ads but they
flipped it so that now it's you have to

you have to buy the youtube premium
because they figured they would make
more money that way i guess
ah yeah i just searched ads
maybe we can confer with the youtube uh
terms of service or whatever but if it
just recently changed then that totally
could be
and okay good to keep in mind stephen
and that also can be where having a
non-profit status can help
because um they will if they see you
have those values and
you are registered then they often can
give you a
either a free or much reduced access to
some of those services
so i just looked at the youtube
monetization
and it says as a youtube partner you'll
be able to earn money from your videos
etc etc
and it requires 1 000 subscribers
and uh 4 000 watch hours we're not at
either of those so we have no
levers maybe when the channel is
eligible it can be
done or not but um we'll figure it out
and of course have a spirit of cognitive
security and attentional regime
engineering and
not having ads when we don't need them
to be there
and eventually working on avenues to
self-sufficiency
in a way where we won't be at the behest
of any other platform
but we could have some more subscribers

that'd be cool

dean and then anyone else i just think

it's interesting because

i don't think search engine optimization

is active inference

but what would happen if we got to that

those minimum numbers in those

thresholds right

yep well there's a lot on our channel

and um it'd be awesome to reach out to

more

people and let the search engines make

it what they may

because we'll be optimizing for

something else

so this closing slide is just again

people can raise their hand if they want

to add

anything just what we'd like to learn

more about and how we want to apply

questions that every expert in the field

will tell you that they are curious

about as well

and beginners too and then the questions

that we ask

usually in a more paper discussion

context

what does a good understanding of our

lab enable

since we're discovering what kind of lab

we are and creating what kind of lab we

are

what would it mean if we understood that

what

would be the predictions and

implications of different understandings

what do we know now that we didn't know

two hours ago

um what are some next steps for
free energy principle and active
inference the ideas or the community
tools too and then what are the goals of
our
research and of our system of interest
which is the lab as organizers
and then the participants as the lab
so cool stuff and um
thanks everyone for participating we
hope that
you stick around share the information
with anyone else
and um yeah if there's no other comments
we'll end it there
awesome